Chapter 10 — Electromagnetism

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. Force on a current-carrying conductor is:

(a) BIL cos θ **(b)** BIL sin θ **(c)** BI/L **(d)** BIL tan θ

2. The direction of force on a conductor is given by:

(a) Ohm's law **(b)** Fleming's left-hand rule **(c)** right-hand rule **(d)** Lenz's law

3. One tesla equals:

(a) Wb m^2 **(b)** J C^{-1} **(c)** N A m **(d)** N A^{-1} m^{-1}

4. Magnetic flux is:

(a) BA cos θ **(b)** A/B **(c)** BA sin θ **(d)** B/A

5. SI unit of magnetic flux is:

(a) farad **(b)** tesla **(c)** weber **(d)** henry

6. Magnetic flux density is measured in:

(a) volt **(b)** tesla **(c)** weber **(d)** ampere

7. Flux linkage for N turns is:

(a) Φ/N **(b)** NΦ **(c)** N + Φ **(d)** N/Φ

8. Force on a moving charge is:

(a) qvB sin θ **(b)** qE sin θ **(c)** qvB cos θ **(d)** mvB

9. A magnetic force does work on a moving charge:

(a) always **(b)** never **(c)** only when parallel **(d)** only when antiparallel

10. A charge moving perpendicular to uniform B follows:

(a) ellipse **(b)** circle **(c)** straight line **(d)** parabola

11. Radius of charged particle path is:

(a) mv/qB **(b)** qB/mv **(c)** q/mvB **(d)** mvB/q

12. Radius is directly proportional to:

(a) charge **(b)** field strength **(c)** momentum **(d)** speed only

13. A velocity selector passes particles with:

(a) v = E + B **(b)** v = B/E **(c)** v = E/B **(d)** v = EB

14. In a velocity selector, electric force is:

(a) qB/E **(b)** qE **(c)** qvB **(d)** qE/B

15. Motional emf for rod moving perpendicular to B is:

(a) vB/L **(b)** BL/v **(c)** v/B L **(d)** vBL

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16. Faraday's law is:

(a) ε = Φ/Nt **(b)** ε = −NΔΦ/Δt **(c)** ε = NΦt **(d)** ε = NΔt/ΔΦ

17. The minus sign in Faraday's law represents:

(a) Gauss's law **(b)** Lenz's law **(c)** Ohm's law **(d)** Coulomb's law

18. Lenz's law is a consequence of conservation of:

(a) flux only **(b)** energy **(c)** momentum only **(d)** charge only

19. Induced emf increases with:

(a) zero turns **(b)** constant flux **(c)** faster flux change **(d)** slower flux change

20. A magnet falls slowly through a copper tube due to:

(a) gravity decreasing **(b)** magnetic monopoles **(c)** air pressure **(d)** induced currents opposing motion

21. Ferrofluids contain:

(a) only iron filings **(b)** only oil **(c)** nanometre magnetic particles **(d)** only water

22. Ferrofluids form spikes due to:

(a) electric field only **(b)** sound **(c)** magnetic field **(d)** gravity only

23. A seismometer uses:

(a) only light **(b)** electromagnetic induction **(c)** only heat **(d)** only gravity

24. In a seismometer, induced emf is proportional to:

(a) velocity of ground motion **(b)** displacement only **(c)** mass only **(d)** temperature

25. A stationary charge in magnetic field experiences:

(a) qvB **(b)** maximum force **(c)** qE **(d)** no magnetic force

26. A current-carrying wire parallel to B experiences:

(a) maximum force **(b)** zero force **(c)** BIL/2 **(d)** BIL

27. A current-carrying wire perpendicular to B experiences:

(a) maximum force **(b)** BIL/2 **(c)** zero force **(d)** BIL/4

28. If current is doubled, force on conductor becomes:

(a) double **(b)** half **(c)** four times **(d)** unchanged

29. If length is doubled, force on conductor becomes:

(a) unchanged **(b)** double **(c)** half **(d)** four times

30. If magnetic field is doubled, force on conductor becomes:

(a) double **(b)** four times **(c)** half **(d)** unchanged

31. The magnetic flux through a coil is zero when field is:

(a) perpendicular to plane **(b)** at 45° **(c)** parallel to plane of coil **(d)** maximum

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32. The magnetic flux is maximum when field is:

(a) at 30° **(b)** parallel to plane **(c)** zero **(d)** perpendicular to plane of coil

33. A charged particle moving parallel to B experiences:

(a) maximum force **(b)** qE **(c)** qvB **(d)** no force

34. A charged particle moving antiparallel to B experiences:

(a) maximum force **(b)** no force **(c)** qvB **(d)** qE

35. Kinetic energy of a charge in uniform magnetic field:

(a) becomes zero **(b)** decreases **(c)** remains constant **(d)** increases

36. A proton and electron with same speed in same B have:

(a) no deflection **(b)** same radius always **(c)** opposite deflection **(d)** same deflection

37. For same mass and speed, slower particle in B is deflected:

(a) same **(b)** not at all **(c)** less **(d)** more

38. A single moving proton:

(a) produces no field **(b)** produces magnetic field **(c)** produces only electric field **(d)** cannot produce field

39. Magnetic flux can be zero even with field if:

(a) field is strong **(b)** current flows **(c)** coil has many turns **(d)** coil plane is parallel to field

40. To induce emf, flux linkage must:

(a) remain constant **(b)** be zero always **(c)** be maximum always **(d)** change

41. A coil in x–y plane with B along y should rotate about:

(a) any axis always **(b)** y-axis **(c)** z-axis only **(d)** x-axis

42. If area and B change but flux linkage stays constant, induced emf is:

(a) BIL **(b)** maximum **(c)** zero **(d)** NΦ

43. Electrical energy in a bulb from moving rod comes from:

(a) magnetic field energy only **(b)** heat only **(c)** chemical energy **(d)** mechanical work

44. Sensitive instruments are shielded from stray magnetic fields by:

(a) wood **(b)** plastic **(c)** soft iron enclosure **(d)** copper only

45. Fleming’s left-hand rule: first finger gives:

(a) force **(b)** velocity **(c)** current **(d)** field

46. Fleming’s left-hand rule: second finger gives:

(a) field **(b)** force **(c)** flux **(d)** current

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47. Fleming’s left-hand rule: thumb gives:

(a) flux **(b)** current **(c)** field **(d)** force

48. The unit N A^{-1} m^{-1} is called:

(a) tesla **(b)** weber **(c)** volt **(d)** coulomb

49. The unit Wb is called:

(a) henry **(b)** weber **(c)** tesla **(d)** farad

50. A velocity selector uses:

(a) parallel E and B **(b)** only E **(c)** perpendicular E and B **(d)** only B

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Answer Key with Brief Reasons

1. Answer: (b) BIL sin θ — F = BIL sin θ.

2. Answer: (b) Fleming's left-hand rule — Left-hand rule gives motor force.

3. Answer: (d) N A^{-1} m^{-1} — B = F/IL.

4. Answer: (a) BA cos θ — Φ = B·A.

5. Answer: (c) weber — Φ is measured in Wb.

6. Answer: (b) tesla — B is flux per unit area.

7. Answer: (b) NΦ — Total linked flux.

8. Answer: (a) qvB sin θ — Magnetic Lorentz force.

9. Answer: (b) never — Force is perpendicular to velocity.

10. Answer: (b) circle — Magnetic force is centripetal.

11. Answer: (a) mv/qB — r = mv/qB.

12. Answer: (c) momentum — r = p/qB.

13. Answer: (c) v = E/B — Electric and magnetic forces balance.

14. Answer: (b) qE — Electric force independent of speed.

15. Answer: (d) vBL — ε = vBL.

16. Answer: (b) ε = −NΔΦ/Δt — Induced emf from flux change.

17. Answer: (b) Lenz's law — Opposition to change.

18. Answer: (b) energy — Induced current opposes cause.

19. Answer: (c) faster flux change — ε ∝ ΔΦ/Δt.

20. Answer: (d) induced currents opposing motion — Lenz's law braking.

21. Answer: (c) nanometre magnetic particles — Suspension of magnetic nanoparticles.

22. Answer: (c) magnetic field — Particles align along field.

23. Answer: (b) electromagnetic induction — Relative motion induces emf.

24. Answer: (a) velocity of ground motion — Faraday's law depends on rate.

25. Answer: (d) no magnetic force — v = 0 gives F = 0.

26. Answer: (b) zero force — sin 0° = 0.

27. Answer: (a) maximum force — sin 90° = 1.

28. Answer: (a) double — F ∝ I.

29. Answer: (b) double — F ∝ L.

30. Answer: (a) double — F ∝ B.

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31. **Answer: (c) parallel to plane of coil** — Normal perpendicular to B.
32. **Answer: (d) perpendicular to plane of coil** — Normal parallel to B.
33. **Answer: (d) no force** — sin 0° = 0.
34. **Answer: (b) no force** — sin 180° = 0.
35. **Answer: (c) remains constant** — Magnetic force does no work.
36. **Answer: (c) opposite deflection** — Opposite charges.
37. **Answer: (d) more** — r ∝ v, smaller radius means more curvature.
38. **Answer: (b) produces magnetic field** — Moving charge is current.
39. **Answer: (d) coil plane is parallel to field** — No perpendicular component.
40. **Answer: (d) change** — Faraday's law.
41. **Answer: (d) x-axis** — Rotation must change flux.
42. **Answer: (c) zero** — No change in flux.
43. **Answer: (d) mechanical work** — Work moves rod against magnetic force.
44. **Answer: (c) soft iron enclosure** — High permeability shields.
45. **Answer: (d) field** — Field, current, force.
46. **Answer: (d) current** — Current direction.
47. **Answer: (d) force** — Motion/force direction.
48. **Answer: (a) tesla** — Magnetic flux density.
49. **Answer: (b) weber** — Magnetic flux.
50. **Answer: (c) perpendicular E and B** — Forces oppose.

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